

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-2800905})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 27 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}906 + \mathbf{Z}(483 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 2800905 = 0$, of discriminant -11203620 . Class group invariants $(10 \ 10 \ 10)$, class number 1000; the three stored generator ideals (HNF) are $\begin{pmatrix} 426 & 51 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1255 & 755 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 13 & 7 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (8s - 33869)y^2 + (340s - 648684)y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216} \right) s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $13070819502 = 2 \cdot 3^4 \cdot 13 \cdot 179 \cdot 34673$:

$$\begin{pmatrix} 484104426 & 375659442 & 231314970 & 20784355 & 196407038 & 443935002 & 450125258 & 456136876 & 450734961 & 164151061 \\ 0 & 3 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 3 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 130 factors with signed exponents of absolute value up to 61432318; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (8s - 33869)y^2 + (340s - 648684)y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10587476640 = 2^5 \cdot 3 \cdot 5 \cdot 13 \cdot 1696711$:

$$\begin{pmatrix} 661717290 & 304571440 & 481114626 & 490455262 & 495451507 & 319162329 & 161602581 & 601870900 & 618810446 & 621630709 \\ 0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 27582645; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 3 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (8s - 33869)y^2 + (340s - 648684)y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2099627642060 = 2^2 \cdot 5 \cdot 13 \cdot 2789 \cdot 2895479$:

$$\begin{pmatrix} 1049813821030 & 734127432014 & 67173029314 & 959389486830 & 57053121859 & 57785365490 & 392561043258 & 895986675712 & 841915140590 & 740512538666 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 133 factors with signed exponents of absolute value up to 63178993; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $412192614796 = 2^2 \cdot 107 \cdot 963066857$:

$$\begin{pmatrix} 206096307398 & 164520907200 & 126567181928 & 49396419060 & 40840969302 & 9830235395 & 196983202531 & 175469218589 & 11327597003 & 146052724215 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 428347; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $247528120320 = 2^{13} \cdot 3^5 \cdot 5 \cdot 13 \cdot 1913$:

$$\begin{pmatrix} 2984280 & 1860624 & 2148252 & 1468916 & 1851016 & 979330 & 635075 & 2060362 & 378920 & 2299871 \\ 0 & 24 & 18 & 0 & 9 & 0 & 2 & 5 & 10 & 7 \\ 0 & 0 & 6 & 4 & 5 & 2 & 4 & 1 & 0 & 2 \\ 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 6 & 0 & 2 & 2 & 4 & 1 \\ 0 & 0 & 0 & 0 & 0 & 12 & 0 & 10 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 159 factors with signed exponents of absolute value up to 18128; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\frac{y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000}{2128000}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right) s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1633673869360 = 2^4 \cdot 5 \cdot 13 \cdot 67^2 \cdot 349931$:

$$\begin{pmatrix} 3047899010 & 641341554 & 683872202 & 2867786190 & 1993074766 & 2128589928 & 2190478923 & 1411705703 & 1734308691 & 2031171313 \\ 0 & 134 & 123 & 10 & 132 & 116 & 0 & 0 & 50 & 83 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 164 factors with signed exponents of absolute value up to 1065502; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 123)y^3 + (10s + 24508)y^2 + (14s - 189214)y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $244984828968 = 2^3 \cdot 3^3 \cdot 11^3 \cdot 113 \cdot 7541$:

$$\begin{pmatrix} 1237297116 & 1107664602 & 373802134 & 673928382 & 599870118 & 19670863 & 259627519 & 441083314 & 19113414 & 997088270 \\ 0 & 66 & 21 & 51 & 33 & 42 & 15 & 33 & 45 & 49 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 2 & 2 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 113 factors with signed exponents of absolute value up to 30412; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 123)y^3 + (10s + 24508)y^2 + (14s - 189214)y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7544716245 = 3^6 \cdot 5 \cdot 11 \cdot 188171$:

$$\begin{pmatrix} 93144645 & 77474589 & 31336536 & 33924207 & 1400637 & 39112834 & 59202031 & 87371245 & 8148687 & 9424261 \\ 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 3 & 0 & 1 & 2 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 3 & 1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 116 factors with signed exponents of absolute value up to 113726; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 123)y^3 + (10s + 24508)y^2 + (14s - 189214)y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $583425695160 = 2^3 \cdot 3^4 \cdot 5 \cdot 7^2 \cdot 11^4 \cdot 251$:

$$\begin{pmatrix} 579810 & 390654 & 439978 & 400906 & 157986 & 129878 & 22438 & 118162 & 331209 & 183238 \\ 0 & 66 & 0 & 0 & 27 & 30 & 30 & 45 & 42 & 7 \\ 0 & 0 & 154 & 121 & 33 & 121 & 105 & 6 & 6 & 137 \\ 0 & 0 & 0 & 33 & 6 & 8 & 4 & 3 & 12 & 31 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 106 factors with signed exponents of absolute value up to 8752; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 62)y^3 + (-7s - 11955)y^2 + (6s - 75342)y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 120y^8 - 23662y^7 + 2701885y^6 - 38681682y^5 + 32 \\ & 5625238y^4 + 7962666480y^3 - 23477418216y^2 + 132301507680y + 39 \\ & 56203667520 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $340301732544 = 2^6 \cdot 3^3 \cdot 241 \cdot 817153$:

$$\begin{pmatrix} 2363206476 & 1646932974 & 1685191470 & 395697078 & 1588385436 & 1086286908 & 793139898 & 255832331 & 1655470358 & 1595617503 \\ 0 & 3 & 1 & 2 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & 4 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 1003112; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 62)y^3 + (-7s - 11955)y^2 + (6s - 75342)y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 120y^8 - 23662y^7 + 2701885y^6 - 38681682y^5 + 32 \\ & 5625238y^4 + 7962666480y^3 - 23477418216y^2 + 132301507680y + 39 \\ & 56203667520 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1655028426480 = 2^4 \cdot 3 \cdot 5 \cdot 11^2 \cdot 13 \cdot 4383949$:

$$\begin{pmatrix} 18807141210 & 1149115242 & 9794350704 & 3804712500 & 13427970918 & 4275596576 & 15626843533 & 9182443815 & 15547433924 & 9395912319 \\ 0 & 11 & 2 & 0 & 1 & 0 & 5 & 5 & 9 & 1 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 919270; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 62)y^3 + (-7s - 11955)y^2 + (6s - 75342)y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 120y^8 - 23662y^7 + 2701885y^6 - 38681682y^5 + 32 \\ & 5625238y^4 + 7962666480y^3 - 23477418216y^2 + 132301507680y + 39 \\ & 56203667520 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $97407492600 = 2^3 \cdot 3 \cdot 5^2 \cdot 11^2 \cdot 1341701$:

$$\begin{pmatrix} 4870374630 & 1915556880 & 3019615650 & 1976487432 & 1779492132 & 3279218472 & 4006393506 & 559354305 & 543373656 & 3289497063 \\ 0 & 10 & 4 & 6 & 4 & 2 & 0 & 7 & 7 & 4 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 130 factors with signed exponents of absolute value up to 1242228; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (1 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 121)y^3 + (16s + 3420)y^2 + (-180s - 305964)y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 233y^8 + 7566y^7 + 2183098y^6 - 84618768y^5 + 179 \\ & 9091024y^4 - 16290272448y^3 + 159330584736y^2 - 788885868672y + \\ & 4526813143296 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216} \right) s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $233867734740 = 2^2 \cdot 3^2 \cdot 5 \cdot 23 \cdot 761 \cdot 74231$:

$$\begin{pmatrix} 38977955790 & 22161669858 & 1286145072 & 18112470847 & 32103976277 & 4808421144 & 27405158875 & 17252601535 & 14450236729 & 30371689007 \\ 0 & 6 & 0 & 2 & 3 & 5 & 0 & 3 & 3 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 129 factors with signed exponents of absolute value up to 38956; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (1 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 121)y^3 + (16s + 3420)y^2 + (-180s - 305964)y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 233y^8 + 7566y^7 + 2183098y^6 - 84618768y^5 + 179 \\ & 9091024y^4 - 16290272448y^3 + 159330584736y^2 - 788885868672y + \\ & 4526813143296 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $22894558920 = 2^3 \cdot 3^2 \cdot 5 \cdot 17 \cdot 607 \cdot 6163$:

$$\begin{pmatrix} 1907879910 & 1327076322 & 823966894 & 644951425 & 329851431 & 1220174512 & 1798843377 & 1470964105 & 717074021 & 994683121 \\ 0 & 6 & 4 & 5 & 1 & 0 & 5 & 0 & 1 & 0 \\ 0 & 0 & 2 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 243599; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 121)y^3 + (16s + 3420)y^2 + (-180s - 305964)y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 233y^8 + 7566y^7 + 2183098y^6 - 84618768y^5 + 179 \\ & 9091024y^4 - 16290272448y^3 + 159330584736y^2 - 788885868672y + \\ & 4526813143296 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2154434142720 = 2^9 \cdot 3 \cdot 5 \cdot 17 \cdot 1861 \cdot 8867$:

$$\begin{pmatrix} 16831516740 & 15024422532 & 7067870470 & 12361579048 & 12286250382 & 13379749198 & 9207865976 & 15603721634 & 4050201654 & 7121128570 \\ 0 & 4 & 2 & 2 & 2 & 0 & 2 & 1 & 3 & 1 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 129 factors with signed exponents of absolute value up to 100605; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 52y^8 + 967726y^6 + 428473620y^4 + 84564207225y^2 + 13803980202000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $178513655400 = 2^3 \cdot 3^2 \cdot 5^2 \cdot 163 \cdot 608431$:

$$\begin{pmatrix} 2975227590 & 186859530 & 2156152170 & 2723959661 & 1624111463 & 1550589379 & 894969644 & 1402533441 & 1806190805 & 378457270 \\ 0 & 30 & 0 & 29 & 23 & 0 & 17 & 13 & 14 & 26 \\ 0 & 0 & 2 & 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 124585997; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 52y^8 + 967726y^6 + 428473620y^4 + 84564207225y^2 + 13803980202000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $372996494700 = 2^2 \cdot 3 \cdot 5^2 \cdot 23 \cdot 179 \cdot 301997$:

$$\begin{pmatrix} 37299649470 & 12584814375 & 4863013453 & 12888650423 & 33161757620 & 28979049959 & 20349123815 & 3239344032 & 6671544396 & 21285579387 \\ 0 & 5 & 4 & 0 & 4 & 3 & 3 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 147592620; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 52y^8 + 967726y^6 + 428473620y^4 + 84564207225y^2 + 13803980202000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right) s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $55781154432 = 2^7 \cdot 3^2 \cdot 23 \cdot 2105267$:

$$\begin{pmatrix} 290526846 & 233846454 & 14543714 & 211131226 & 203783830 & 231580250 & 96562340 & 91623721 & 172620616 & 111229417 \\ 0 & 6 & 0 & 4 & 1 & 0 & 2 & 0 & 4 & 3 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 155 factors with signed exponents of absolute value up to 8392999; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 19: character $2a$, column e_1

Prescribed character vector $(2 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)y^2 + (-340s - 648684)y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $13070819502 = 2 \cdot 3^4 \cdot 13 \cdot 179 \cdot 34673$:

$$\begin{pmatrix} 484104426 & 375659442 & 231314970 & 20784355 & 196407038 & 443935002 & 450125258 & 456136876 & 450734961 & 164151061 \\ 0 & 3 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 3 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 150967808; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 20: character $2a$, column e_2

Prescribed character vector $(2 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)y^2 + (-340s - 648684)y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $379155860616 = 2^3 \cdot 3 \cdot 13 \cdot 499 \cdot 2435357$:

$$\begin{pmatrix} 94788965154 & 64360777590 & 1897812564 & 3211914925 & 45867235664 & 8730312770 & 91438839032 & 77122522922 & 12217977932 & 54528951723 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 134 factors with signed exponents of absolute value up to 266216466; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 2 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 21: character $2a$, column e_3

Prescribed character vector $(2 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)y^2 + (-340s - 648684)y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $158406917760 = 2^7 \cdot 3^6 \cdot 5 \cdot 7^2 \cdot 13^2 \cdot 41$:

$$\begin{pmatrix} 223860 & 119406 & 179382 & 86262 & 70458 & 48720 & 98598 & 193694 & 138302 & 143952 \\ 0 & 21 & 12 & 12 & 0 & 3 & 0 & 15 & 6 & 17 \\ 0 & 0 & 78 & 24 & 60 & 66 & 48 & 18 & 54 & 58 \\ 0 & 0 & 0 & 3 & 2 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 363594176; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 22: character $2b$, column e_1

Prescribed character vector $(0 \ 2 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5293226700 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 557 \cdot 10559$:

$$\begin{pmatrix} 176440890 & 5809020 & 87211030 & 83818622 & 7566778 & 30593077 & 86521691 & 5971211 & 18737894 & 35710437 \\ 0 & 15 & 14 & 8 & 8 & 0 & 11 & 0 & 0 & 14 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 158 factors with signed exponents of absolute value up to 1397128; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 23: character $2b$, column e_2

Prescribed character vector $(0 \ 2 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right)s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $213512823864 = 2^3 \cdot 3^2 \cdot 107 \cdot 509 \cdot 54449$:

$$\begin{pmatrix} 17792735322 & 12119088490 & 6549266688 & 12824107860 & 9112035044 & 11884733755 & 17024466190 & 7019531977 & 4967762761 & 7957321765 \\ 0 & 2 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 391; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 24: character $2b$, column e_3

Prescribed character vector $(0 \ 2 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $85548070400 = 2^9 \cdot 5^2 \cdot 13^2 \cdot 71 \cdot 557$:

$$\begin{pmatrix} 20564440 & 152776 & 4253744 & 9992654 & 11512468 & 7676454 & 20314856 & 9135484 & 18860211 & 1563759 \\ 0 & 52 & 28 & 33 & 34 & 22 & 34 & 8 & 42 & 41 \\ 0 & 0 & 10 & 0 & 0 & 9 & 9 & 2 & 7 & 9 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 163 factors with signed exponents of absolute value up to 4221650; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 25: character $2c$, column e_1

Prescribed character vector $(0 \ 0 \ 2)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 123)y^3 + (-10s + 24508)y^2 + (-14s - 189214)y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} &y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ &8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ &5141073216 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216} \right) s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $102545175744 = 2^6 \cdot 3^8 \cdot 11 \cdot 149^2$:

$$\begin{pmatrix} 59004 & 17028 & 20136 & 3336 & 57378 & 38613 & 15177 & 35244 & 15977 & 27605 \\ 0 & 36 & 18 & 6 & 30 & 18 & 22 & 11 & 0 & 0 \\ 0 & 0 & 894 & 66 & 477 & 798 & 191 & 310 & 872 & 844 \\ 0 & 0 & 0 & 6 & 3 & 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 108 factors with signed exponents of absolute value up to 57768; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (4 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 26: character $2c$, column e_2

Prescribed character vector $(0 \ 0 \ 2)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 123)y^3 + (-10s + 24508)y^2 + (-14s - 189214)y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $357730716090 = 2 \cdot 3 \cdot 5 \cdot 11 \cdot 71 \cdot 151 \cdot 101113$:

$$\begin{pmatrix} 357730716090 & 279888645419 & 239627999627 & 114743485313 & 172262182433 & 65940137727 & 4728492971 & 219291936158 & 99122809393 & 154733809858 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 49107; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 27: character $2c$, column e_3

Prescribed character vector $(0 \ 0 \ 2)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 123)y^3 + (-10s + 24508)y^2 + (-14s - 189214)y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2474463940740 = 2^2 \cdot 3^5 \cdot 5 \cdot 11 \cdot 113 \cdot 157 \cdot 2609$:

$$\begin{pmatrix} 45823406310 & 20684903568 & 25290118320 & 35583112656 & 18983639054 & 44214279718 & 26842305973 & 17060040143 & 7532526746 & 37334140205 \\ 0 & 6 & 3 & 3 & 0 & 3 & 3 & 3 & 3 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 106 factors with signed exponents of absolute value up to 11365; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.